

$$n=7$$

$$e^0 = 2n^2$$

$$= 2(7)^2$$

$$= 98$$

orbitals = 49

no. of orbital in any shell = n^2

$n=6$

l	m
0	0
1	-1, 0, +1
2	-2, -1, 0, +1, +2
3	-3, -2, -1, 0, +1, +2, +3
4	-4, -3, -2, -1, 0, +1, +2, +3, +4
5	-5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5

orbital

$n=5$
 $m=0$
 $n=5$

l	m
0	0
1	-1, 0, +1
2	-2, -1, 0, +1, +2
3	-3, -2, -1, 0, +1, +2, +3
4	-4, -3, -2, -1, 0, +1, +2, +3, +4

$m = \pm l$

Electronic configuration $\Rightarrow$

- ① Aufbau principle —
- ② Pauli's exclusion principle
- ③ Hund's rule of max^m multiplicity

① Aufbau principle:

① Distribution of e^- in diff subshell

→ e^- are distributed according to energy level of subshell

→ The subshell having less energy should filled 1st.

→ (n+l) rule → ① subshell having low (n+l) value has less energy.

② for same $(n+l)$ value $\rightarrow$
subshell having low n value
have less energy.

[illegible]

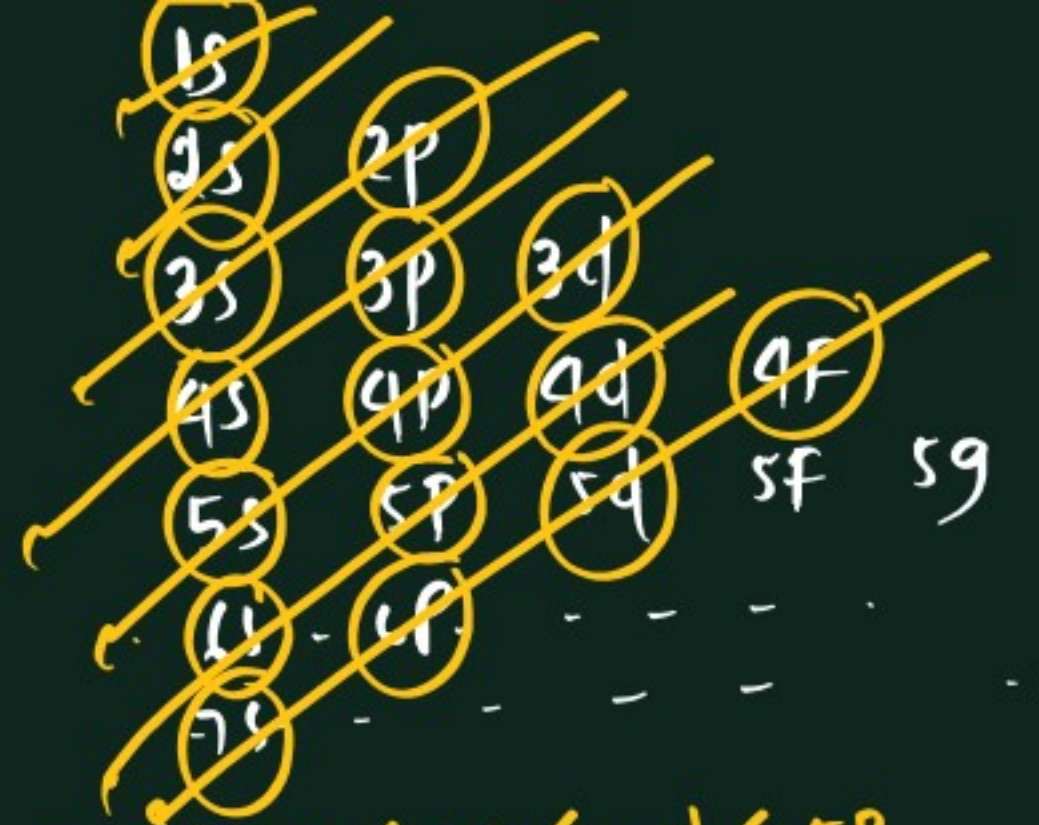
subshell	$n+l$ value
1s	1
2s	2
2p	3 ✓
3s	3 ✓
3p	4
3d	5
4s	4
4p	5
4d	6
4f	7
5s	5

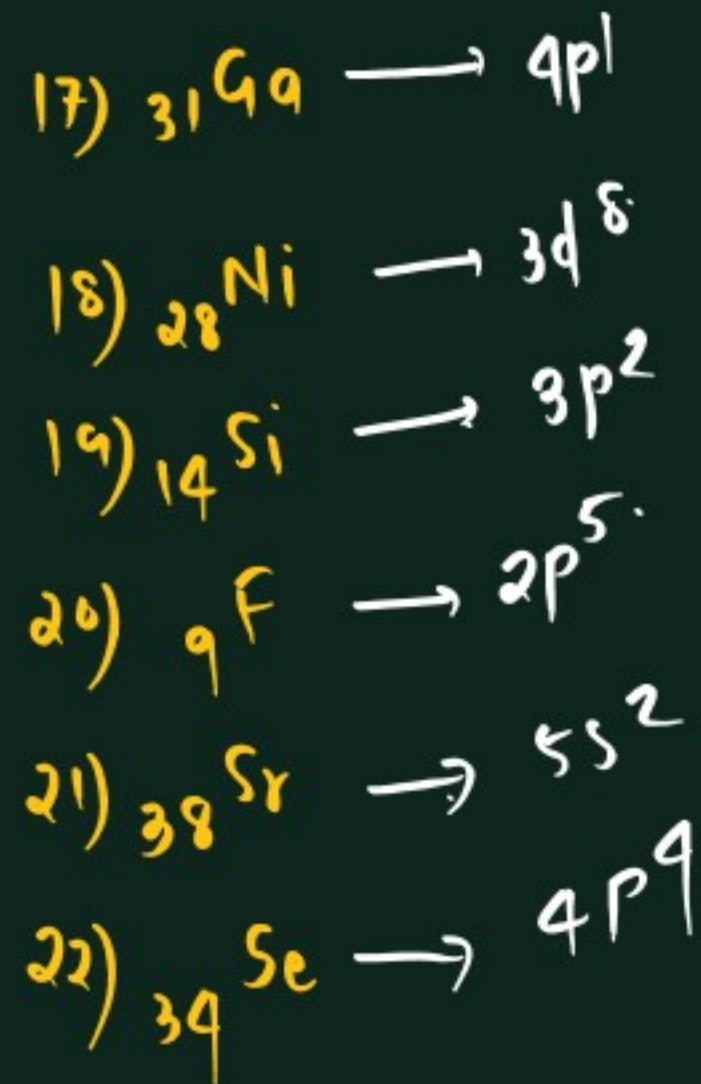
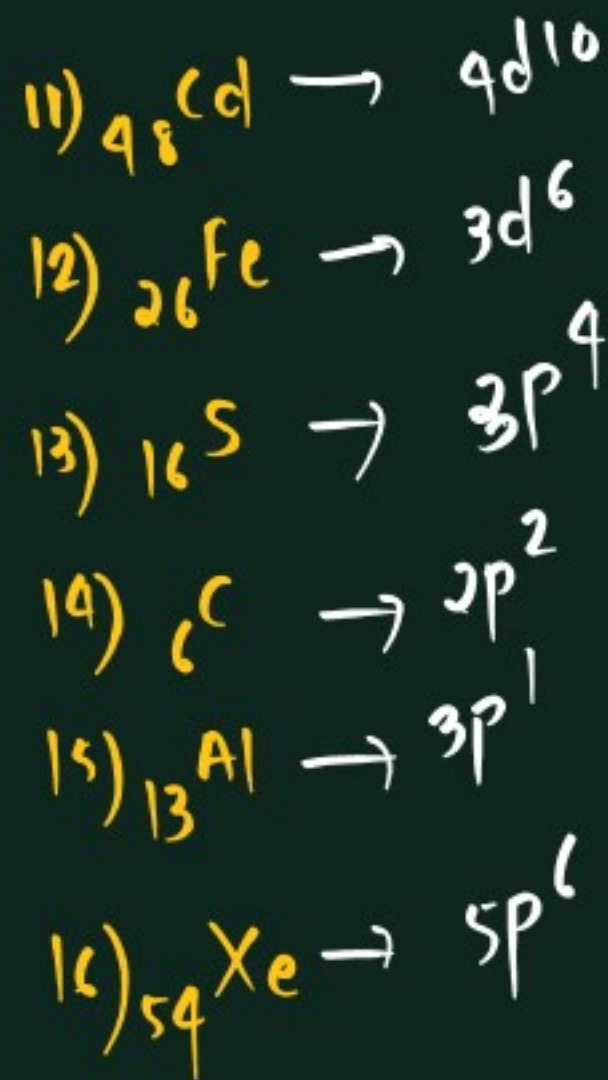
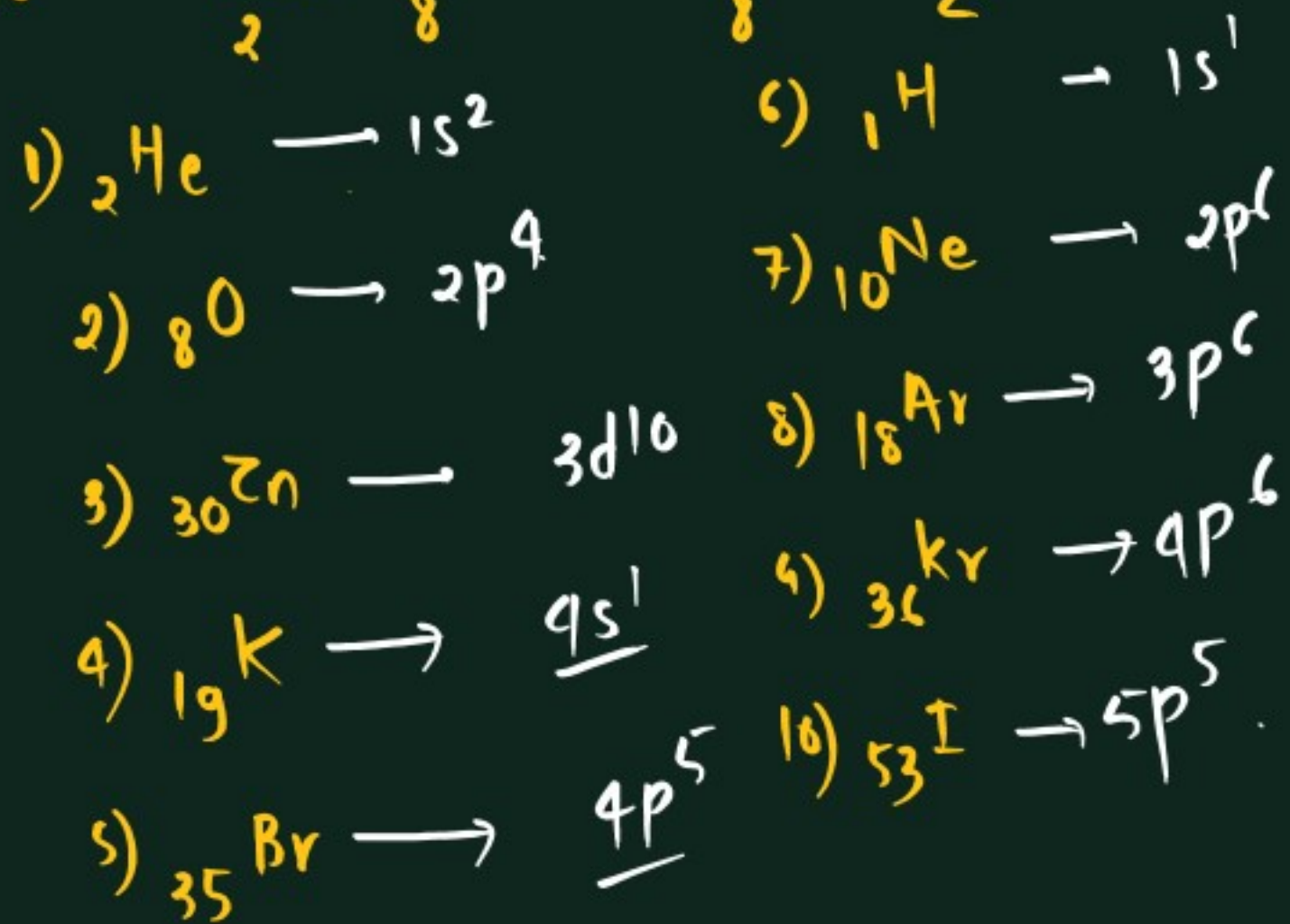
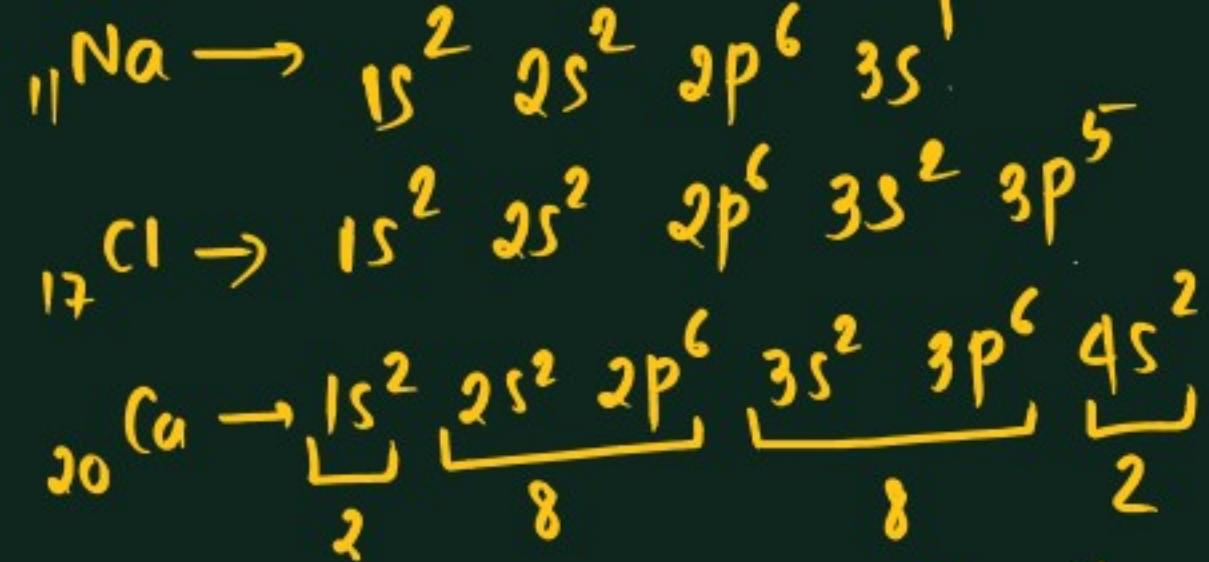
subshell $\rightarrow$ $n+l$ value
 n is not of l

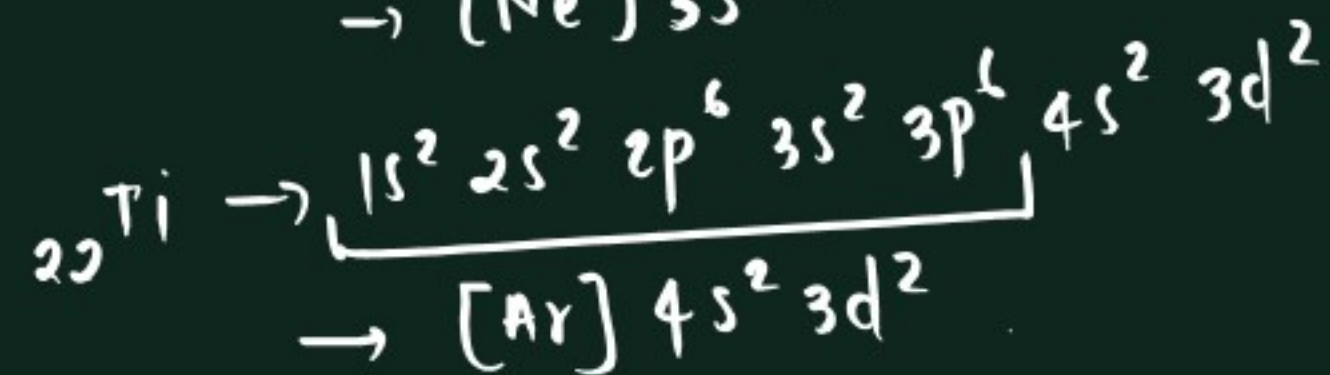
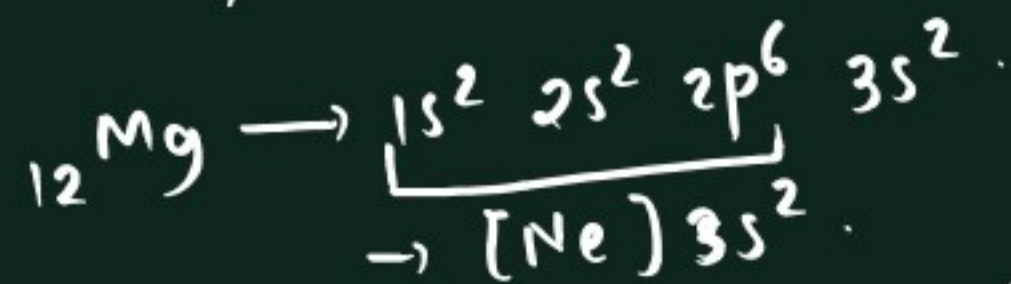
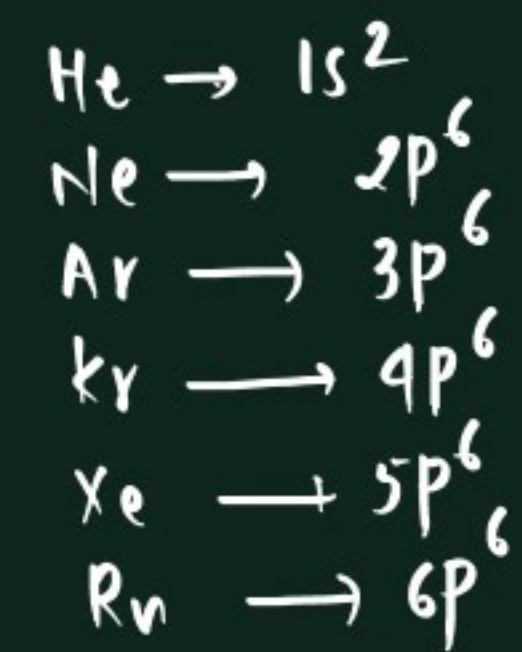
1
 $0 \rightarrow s$
 $1 \rightarrow p$
 $2 \rightarrow d$
 $3 \rightarrow f$

order of energy $\rightarrow$

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p \dots$







② Pauli's exclusion principle $\rightarrow$

$\rightarrow$ distribution of e^- in any orbital

$\rightarrow$ Two e^- present in any orbital always have opposite spin

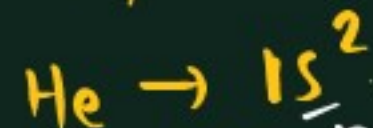
$\rightarrow$ Each orbital represented by $\rightarrow \square$

$\rightarrow \boxed{\uparrow\downarrow} \checkmark$

$\boxed{\uparrow\uparrow} \times$

$\boxed{\downarrow\downarrow} \times$

$\rightarrow$ No two e^- in any orbital will have same set of quantum no. in atom.



	n	l	m	s
1st e^-	1	0	0	$+\frac{1}{2}$
2nd e^-	1	0	0	$-\frac{1}{2}$

3) Hund's rule of max^m multiplicity →
 → Distribution of e^- in orbitals within any subshell.

s →

p → +
|
 $l=1$ -1 0 +1

d → | | | | |
 $l=2$

f → | | | | | | |

$80 - 1s^2 2s^2 2p^4$
↑↓ ↑↓ ↑↓ ↑ ↑

p^4 → ↑↓ ↑ ↑ ✓✓

p^4 → ↑↓ ↑↓ ✗

eg d^7 → ↑↓ ↑↓ ↑ ↑ | |

f^4 → ↑ ↑ ↑ ↑ | | |

	Pauli's	Hund's
1) ↑↓ ↑↓	✓	✗
2) ↑↑ ↑ ↑	✗	✓
3) ↑↓ ↑ ↑	✓	✓
4) ↑↑ ↑	✗	✗
5) ↑↑ ↑ ↑ ↑ ↑	✗	✓

Exceptions →



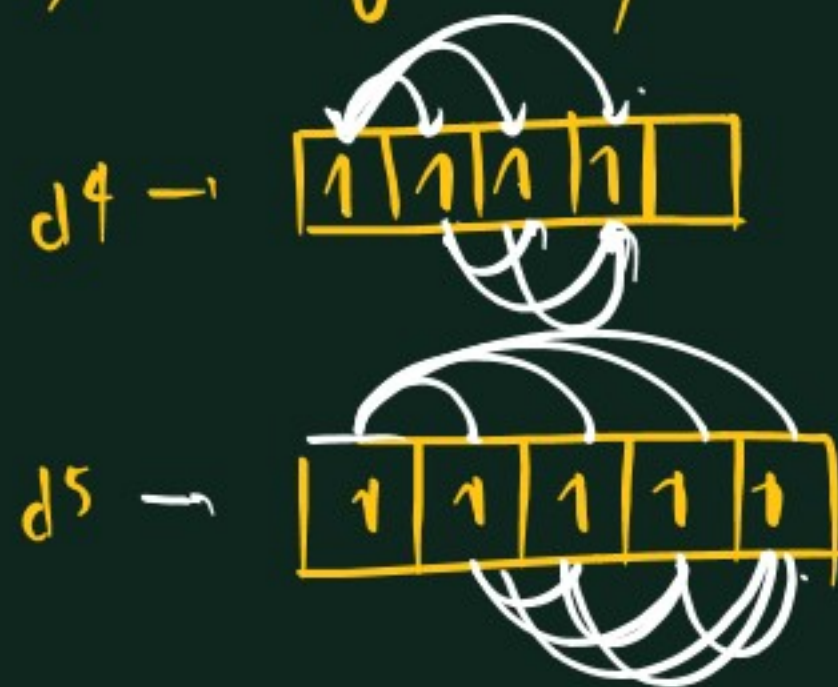
→ The subshell having half filled or fully filled configuration are more stable.

Half filled → p^3, d^5, f^7
Fully filled → s^2, p^6, d^{10}, f^{14}

Extra stability of half filled and fully filled orbitals is due to

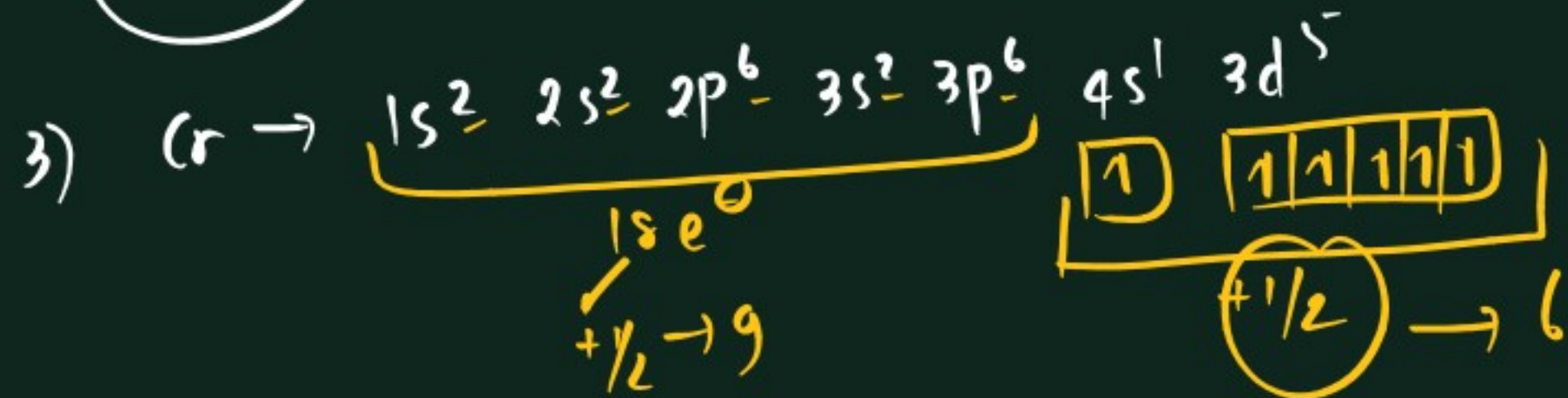
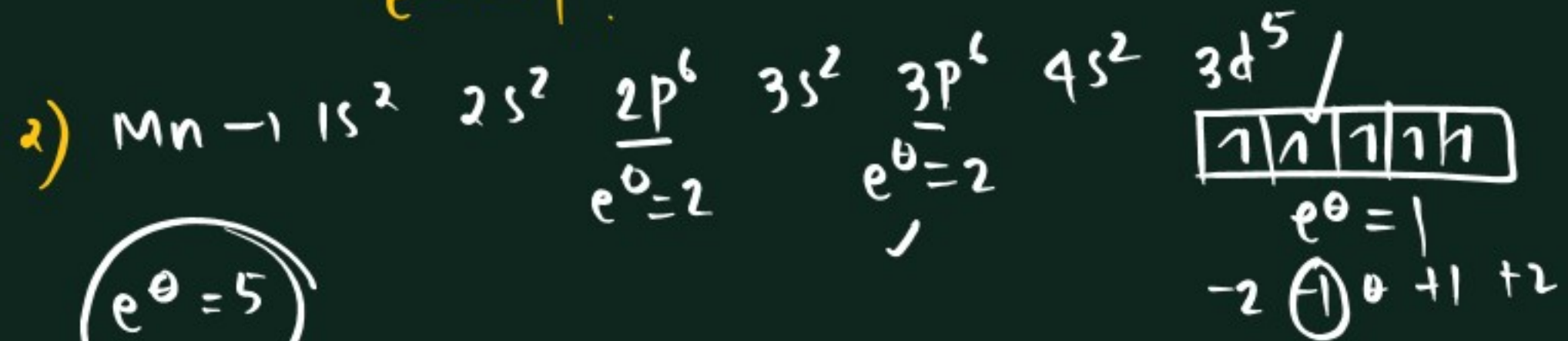
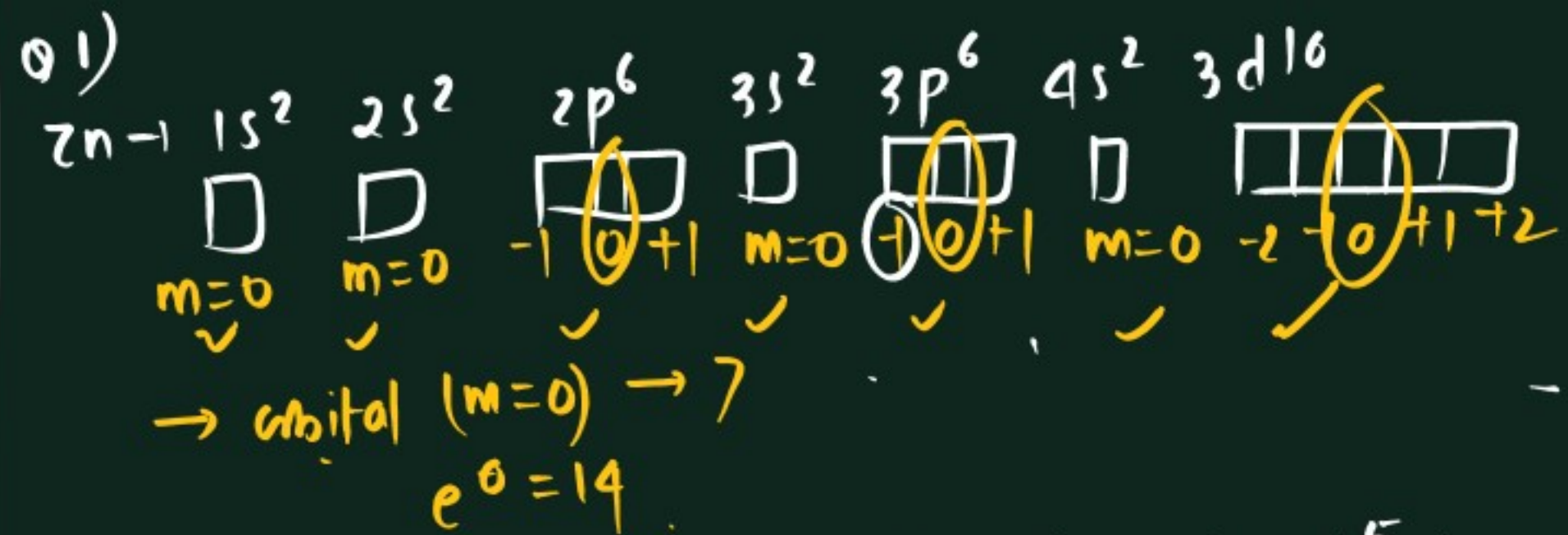
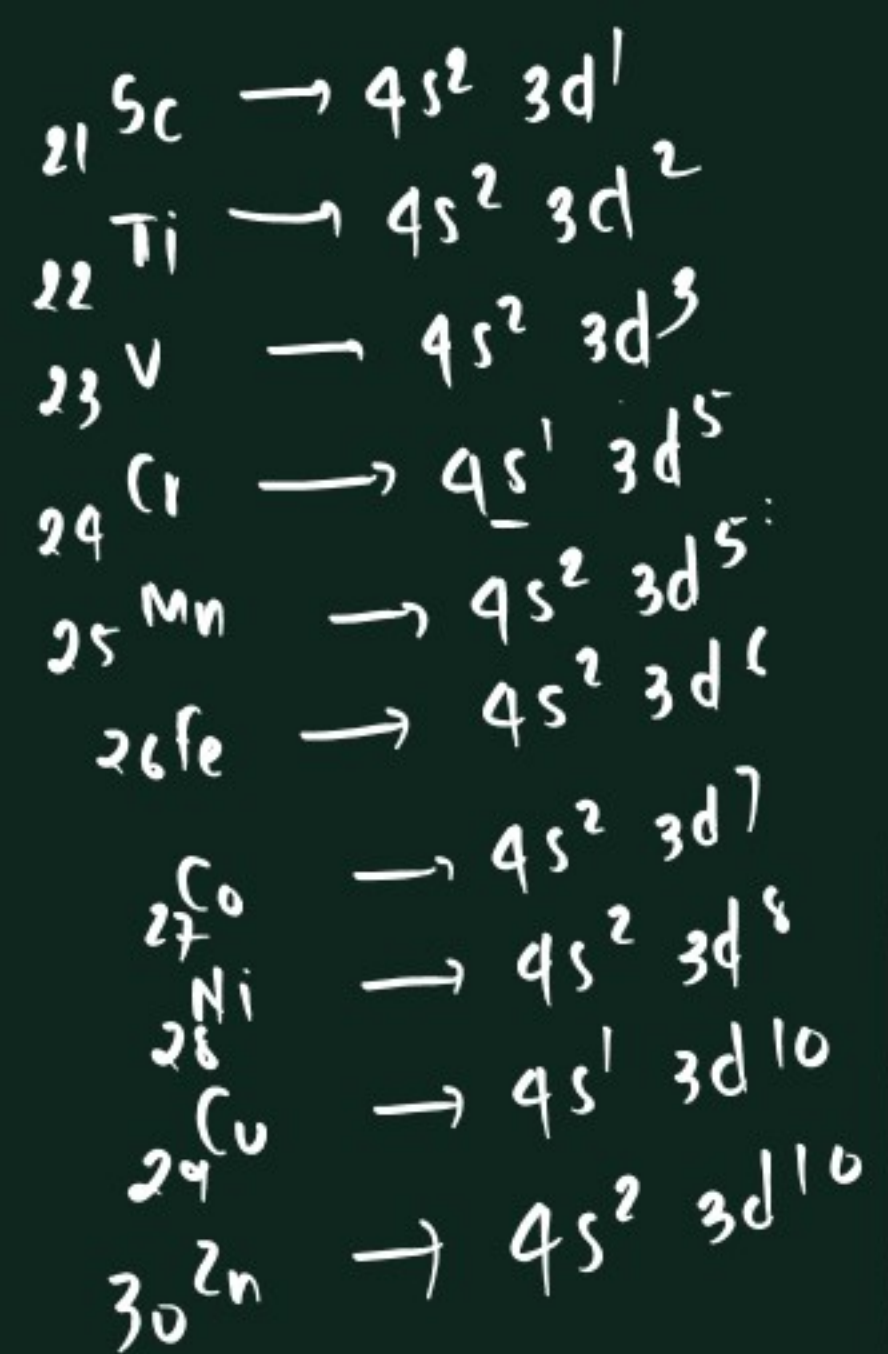
1) Symmetrical distribution of e^-

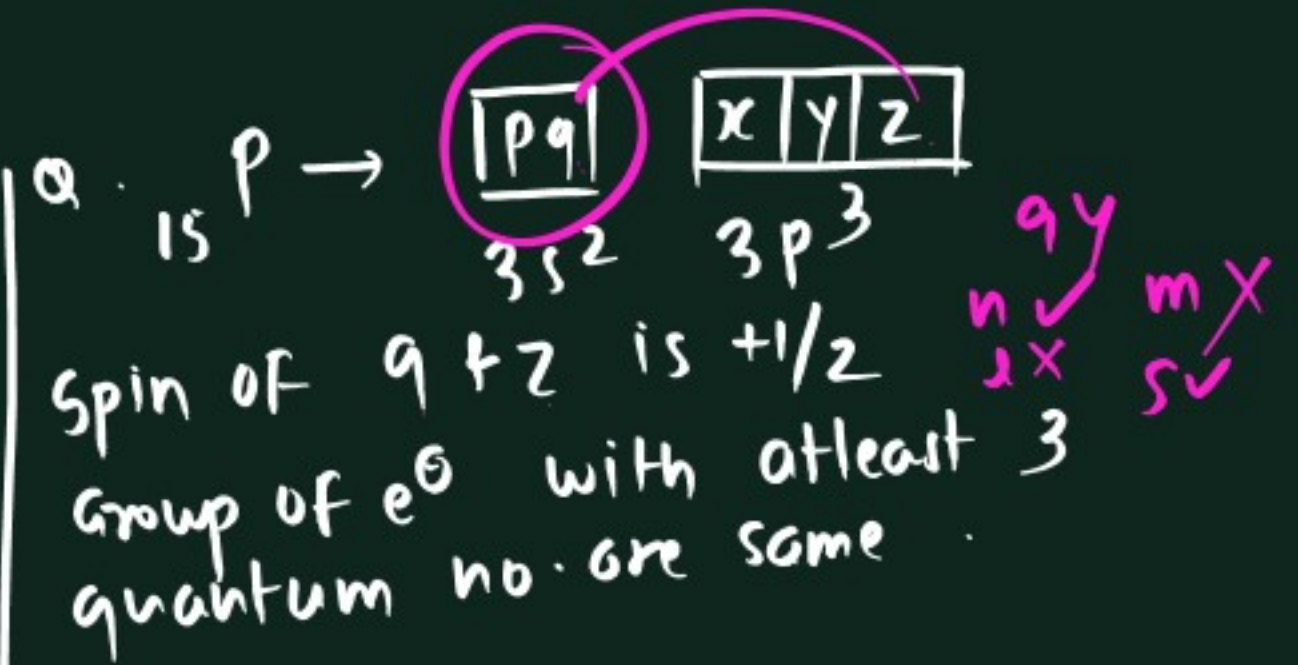
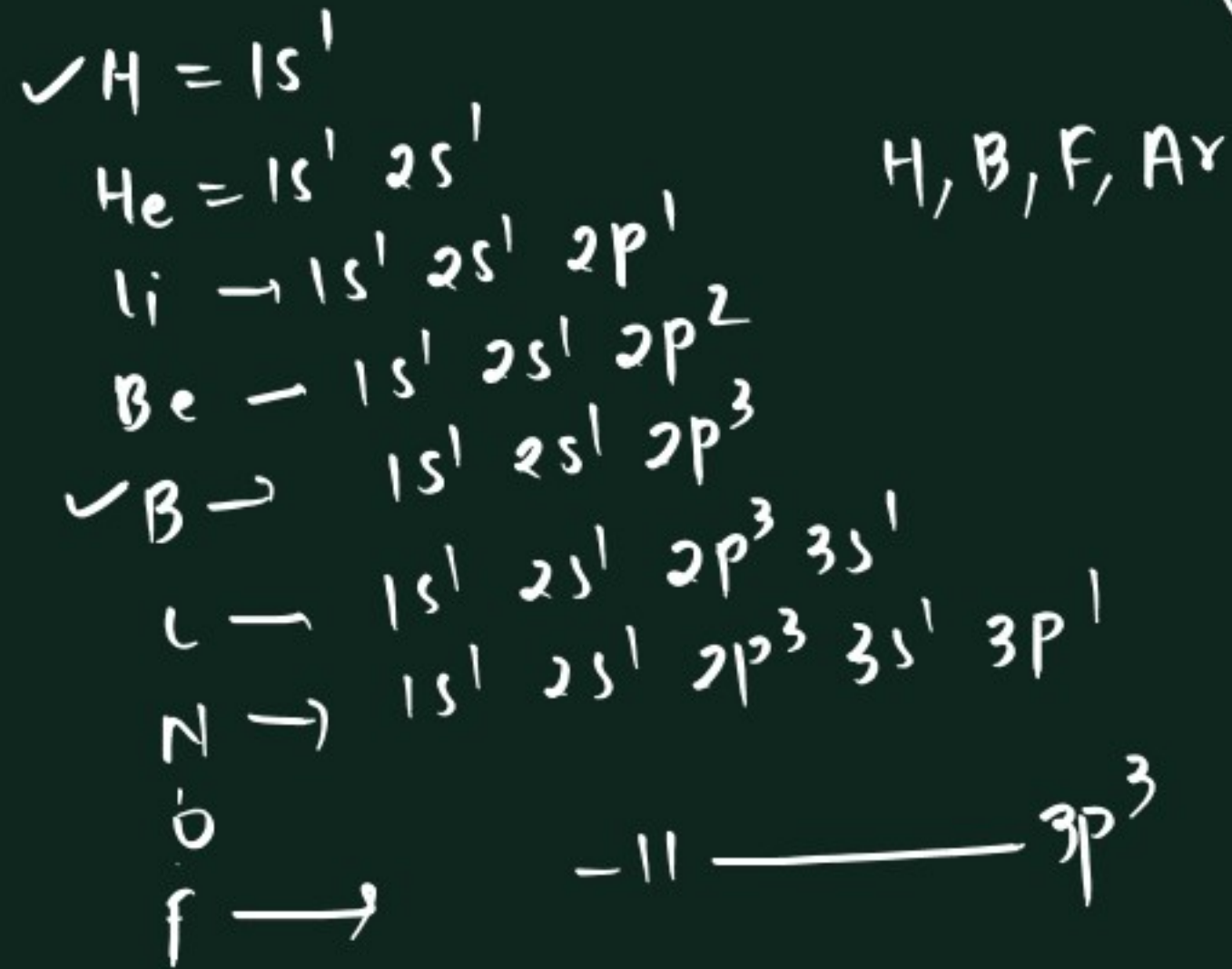
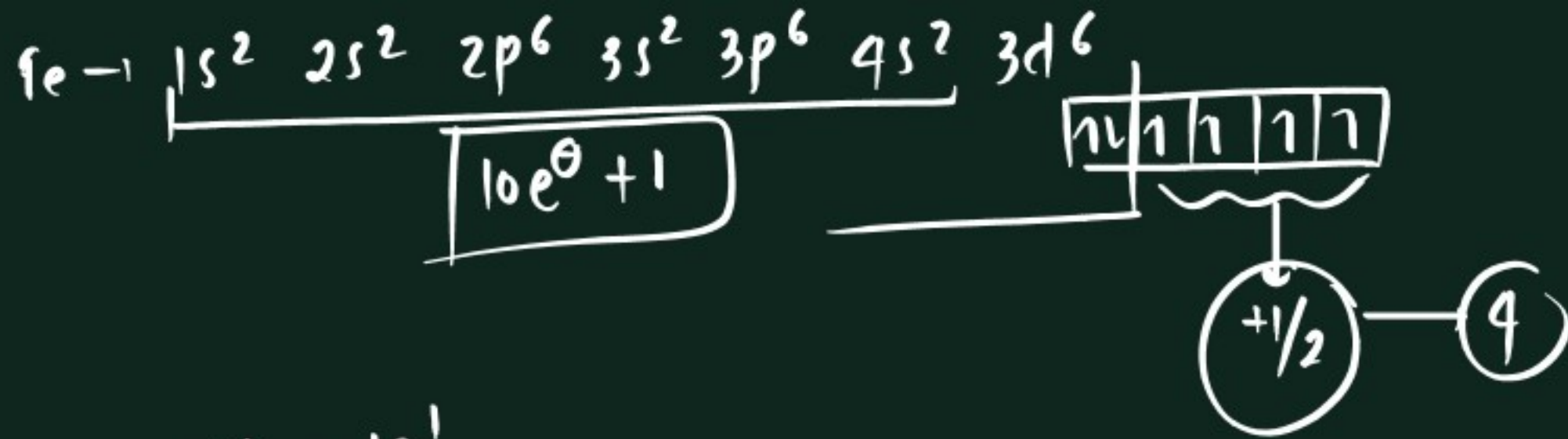
2) Exchange energy →



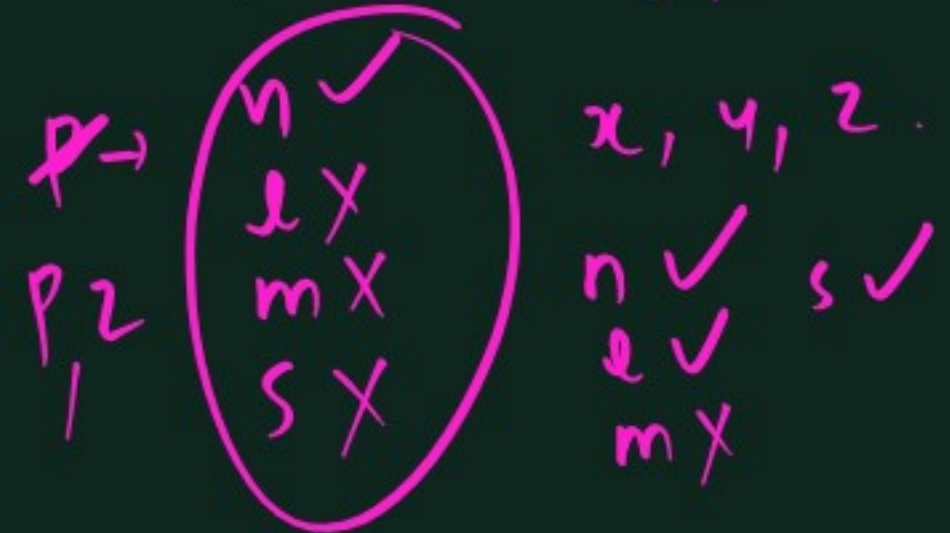
For Notes
website

<https://notesmaker-tuwin.onrender.com>

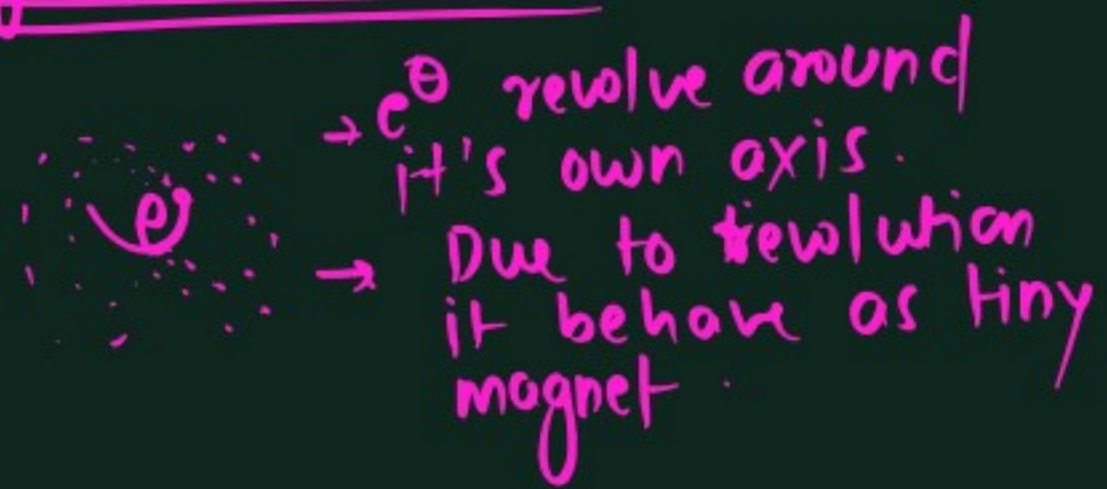




- a) pq $\times$
 b) pq, xyz
 c) pq, xyz, pz $\times$
 d) pq, xyz, qy



Magnetic properties.

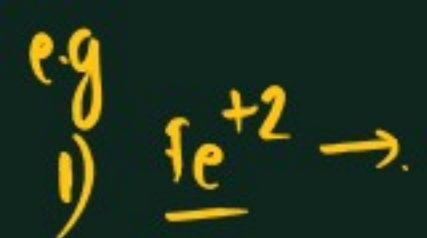


$\uparrow\downarrow$ → paired $e^{\ominus}$
→ magnetic field of both $e^{\ominus}$ will cancel each other

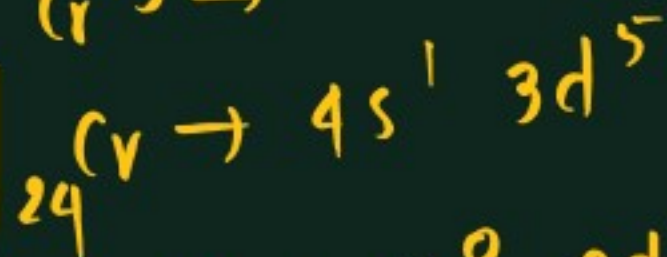
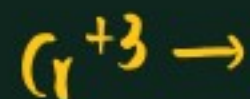
$\uparrow$ → unpair $e^{\ominus}$
→ It will show magnetic behavior.

Paramagnetic	Diamagnetic
→ weakly attracted by mag. field	→ weakly repel by mag. field.
→ unpair $e^{\ominus} \neq 0$ (n)	→ unpair $e^{\ominus} = 0$ (n)
→ B	
Magnetic moment (u) = $\sqrt{n(n+2)}$	B.M ↓ Bohr magneton

e.g



$$\begin{aligned} \rightarrow \mu &= \sqrt{n(n+2)} \\ &= \sqrt{4(4+2)} \\ &= \sqrt{24} \\ &= 4.9 \text{ B.M.} \end{aligned}$$

 $n=3$ Para.

$$\begin{aligned} \mu &= \sqrt{n(n+2)} \\ &= \sqrt{3(3+2)} \\ &= \sqrt{15} \\ &= 3.9 \text{ B.M.} \end{aligned}$$

n	μ
1	1.3
2	2.8
3	3.9
4	4.9
5	5.9

